SOCIAL STATISTICS UNIT

Faculty of Humanities
School of Social Sciences

SOST71032
Network Analysis
2025/26

Course Syllabus

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1 Essential Information

Course convenor

Prof. Philip Leifeld

Lecturers

Prof. Philip Leifeld (philip.leifeld@manchester.ac.uk)
Dr. Yan Wang (yan.wang-5@manchester.ac.uk)

Teaching assistant

Jingyi Kang (jingyi.kang@postgrad.manchester.ac.uk)

Contacts

Questions on any issues beyond course content (schedule, technical problems, assessment submission, etc.): studentsupportpgsoss@manchester.ac.uk

Questions on the course content should be asked in the lectures, practicals, drop-in sessions, and office hours (see below).

Times, Dates, Places

(check Canvas and personal timetables for updates)

Lectures:	Mon, 2-4pm, HBS G6: 9 Mar, 16 Mar, 23 Mar, 20 Apr, 27 Apr. Fri, 2-4pm, HBS G32: 8 May (rescheduled due to bank holiday).
Practicals:	Mon, 4-6pm: 9 Mar, 16 Mar, 23 Mar, 20 Apr, 27 Apr, 11 May. Location: Simon 2.61.
Drop-in sessions:	Thu, 5-6pm, HBS Cordingley lecture theatre: 19 Mar, 26 Mar.

Formative Assessment

1. Descriptive analysis

Release:	Mon 16 Mar, 6pm
Submission deadline:	Mon 30 Mar, 6pm

2. ERGMs

Release:	Mon 13 Apr, 6pm
Submission deadline:	Mon 27 Apr, 6pm

Summative Assessment

Release:	Mon, 11 May, 6pm
Submission deadline:	Mon, 8 Jun, 6pm

Office Hours

To arrange a meeting, please email the corresponding lecturer in advance.

On descriptive network analysis:	Dr. Yan Wang (yan.wang-5@manchester.ac.uk) by appointment.
On network modelling:	Prof. Philip Leifeld (philip.leifeld@manchester.ac.uk) by appointment.

Important: Students must read their university e-mails regularly as important information will be communicated this way. It is sometimes necessary to make changes such as seminar rooms and assessment details and such changes will be communicated by email. Failing to check your emails will not be an acceptable excuse for non-attendance or missed deadlines.

2 Course Content

2.1 Aims

- Present the rationale for statistical network modelling.
- Define network models.
- Introduce key statistical models for network analysis.
- Teach applying statistical modelling to empirical data.

2.2 Learning Outcomes

On completion of this unit successful students will be able to:

Knowledge and Understanding:

- Critically engage with the theoretical foundations of network analysis and use them to formulate empirical questions relevant to network analysis.
- Design and develop network studies.
- Understand the variety of network data.
- Assess the applicability of network-analytical techniques to a given dataset.
- Understand the motivation behind the statistical modelling of networks.
- Critically understand and evaluate network-analytical research, reflect upon related methodology in a theoretically-informed way.
- Understand network-analytical research questions in multidisciplinary contexts, and efficiently operationalise them.

Intellectual skills:

- Critically discuss network-analytical literature applying complex statistical models and identify the most appropriate statistical model for a given research problem.
- Examine network structures using descriptive measures, and statistically model the mechanisms for social network structuring.
- Choose appropriate techniques for network data visualisation.
- Report results of social network analysis in written form.

Practical skills:

- Design and develop tailored network-analytical research projects on a variety of real-world problems.
- Collect, manage, and analyse online and offline datasets, and efficiently approach network data analysis and management.
- Produce state-of-the-art network data visualisations.
- Be proficient in network analysis software.

Transferable skills and personal qualities:

- Develop new or enhanced skills to identify and use diverse social network data and use such data

to inform research projects and interventions in a variety of contexts.
- Understand and mediate multidisciplinary environments and liaise across different intellectual and practical contexts involved in network studies.
- Work collaboratively, both face-to-face and online, on network-analytical projects.
- Accurately and effectively work with numbers and use advanced computational software for network analysis.

3 General Course Readings

All listed readings should be available from the University Main Library:

- Rawlings, C. M., Smith, J. A., Moody, J., and McFarland, D. A. (2023). *Network Analysis. Integrating Social Network Theory, Method, and Application with R.* Cambridge University Press, Cambridge, MA.
- Cranmer, S. J., Desmarais, B. A., and Morgan, J. W. (2020). *Inferential Network Analysis*. Cambridge University Press, Cambridge, MA.
- Lusher, D., Koskinen, J., and Robins, G. (2013). *Exponential Random Graph Models for Social Networks.* Cambridge University Press, New York.

Weekly readings are detailed in a separate document on Canvas: sost71032_weekly_readings.pdf.

4 Course Plan

This is preliminary plan of the course. More information and detail about each week's lecture and workshop will be available on the course website on Canvas.

Week	Lecture	Workshop
1	Network Mechanisms, Dependence, and Configurations	Network data entry and manipulation in R. Computing descriptive measures and visualising networks using statnet, igraph, tidygraph, and ggraph.
2	Exponential Random Graph Models: Specification	Estimating exponential random graph models (ERGM) in R and selecting model terms using the ergm package.
3	Exponential Random Graph Models: Estimation and Sampling	Interpreting ERGM results, evaluating goodness of fit, simulating networks, and diagnosing model degeneracy.
4	Panel Network Models	Estimating temporal exponential random graph models using the btergm package.
5	Relational Event Models	Estimating relational event models using remstimate and related packages.
6	Permutation Approaches and Latent Space Models	Running CUG and QAP tests manually and with the sna package. Estimating latent space and additive and multiplicative latent-factor models using latentnet and amen.

5 Assessment

During assessment students are meant to demonstrate they can tie together conceptual ideas from the literature, understand the research context, handle and visualise data, as well as conduct meaningful descriptive analysis, statistical modelling, and interpretation – to draw reasonable conclusions about the research case. Reports are to include an empirical analysis of the data provided as part of the assignment releases.

Note. Assignments will require elements of practical data analysis. To successfully complete the Assignment, **students must take part in the workshops**, where they will receive training in using network-analytical coding.

5.1 Formative Assessment

These assessments are to prepare students for the summative assessment. Formative assessment will not be marked but commented upon.

Formative Assignment 1: Descriptive analysis

In this assignment, students are asked to describe and visualise a network that is provided using methods and software tools used in the lectures, practicals, and additional materials. The analysis should connect theoretical ideas about the mechanisms in the network to measurable configurations and evaluate these configurations in a descriptive analysis, in preparation for the second formative assignment, which is about statistical modelling. Details will be provided in a separate document on Canvas.

Formative Assignment 2: Exponential random graph modelling

In this assignment, students are asked to estimate exponential random graph models (ERGM) for a network dataset that is provided. Following the first formative assignment, the focus of the second formative assignment will be on selecting configurations and forming a model specification, estimating an ERGM, assessing goodness of fit and diagnosing model degeneracy. A crucial element of the assignment is to adjust the model specification and update the analysis iteratively until the model fits well and theoretical insights can be meaningfully derived. Details will be provided in a separate document on Canvas.

5.2 Summative Assessment

The assignment will be one written essay worth 100% of the total mark.

The word count must not exceed 2000 words.

The essay must include at least:

(1) presentation of research case as an Introduction

(2) network visualisation

(3) descriptive network statistics

(4) textual interpretation of network visualisation and descriptive statistics in the research context

(5) textual justification of selection of network configurations to include in the models based on the

research context and features of the data

(6) table with ERGM model parameters and other conventional statistics

(7) goodness of fit tests

(8) textual interpretation of modelling results accounting for the research context and features of the data.

(9) Conclusion summarising findings overall in the research context.

Details will be provided in a separate document on Canvas.

5.3 Submission of Assignments

Submission of formative assignments is to the TA(s) via email.

Submission of summative assignment:

- You must include an accurate word count on the front page or at the beginning of your report. Failure to do so will lead to an automatic 2-mark deduction. Your word count must include all text in the report but not the bibliography, figures, or tables.
- Assessment criteria are given in Section 9.
- The report must be written using RMarkdown in RStudio. Both the .Rmd source file and the compiled .html or .pdf document must be submitted.
- You must submit your essay by 6pm on the deadline day.
- Full details of how to submit online are available in the 'Assignments' folder in the relevant section on the course Canvas website.
- You must keep a copy of your submission receipt until all work on this course is complete and you have received your final grades.
- Note that our online submission system includes plagiarism detection software. Be sure that you fully understand what plagiarism is and what are its consequences; further details are included in Section 9. If, after reading the guidance, you are at all unsure about what counts as plagiarism, you should contact your Academic Advisor to discuss it.
- If you have technical or organising questions on submission, appeals, resits or similar issues, you are expected to contact student administration at studentsupportpgsoss@manchester.ac.uk. Note that teachers cannot alter marks after announcement, but you can appeal to the corresponding committee, asking student administration about it. Inquiries about submission technical issues, appeals, resits, or mark alteration will not be responded to by the teachers. If you have serious problems submitting on the day, please contact the postgraduate student support urgently.

6 Mitigating Circumstances

Extensions may be granted to students where there are exceptional mitigating circumstances (e.g., strong medical reasons). In such cases a Mitigating Circumstances Form must be completed and submitted to the School (not to the instructors!). Full guidance on mitigating circumstances is available **here**.

7 Feedback

All Social Statistics courses include both formative feedback, which lets you know how you are getting on, and what you could do to improve, and summative feedback, which gives you a mark for your assessed work.

This course uses the following mechanisms for feedback:

- Informal verbal feedback during lectures, practicals, and drop-ins.
- Written formative and summative feedback on assessed coursework, via Canvas.
- Please book an office hour slot by email to let your lecturer know in advance that this is what you want to do.

We're continually working to improve our teaching practices, for that we need your feedback. Towards the end of the semester, you will be asked to fill out a Unit Survey for each of your modules – **please do!** The survey is designed to be very short and easy to fill out, but the results are valuable for our monitoring of teaching quality. We want to hear from you whether your opinion on the course was good, bad, or indifferent.

8 Your Commitment

8.1 Weekly lectures and readings

It is expected that you thoroughly read and prepare the weekly reading assignments. A separate document on Canvas lists the weekly reading assignments. The lecture will then summarise key aspects of the readings, discuss perspectives, and fill in any necessary contents not provided by the readings. Merely attending the lectures without preparing the readings is not sufficient. You are expected to spend considerable time each week on preparation. Please feel free to ask questions about the readings during the lecture. Do not rely on AI summaries. Only close reading in combination with practical application can help you learn the contents thoroughly. During lectures and while reading, please take notes for yourself to support your learning.

8.2 Workshops: Preparation

Workshops are a central part of the course module structure. They provide you with an opportunity to discuss, apply, and enhance your knowledge, and to build confidence in your skills of analysis, comprehension, and presentation. What you will gain from them depends on your preparation and willingness to participate. It is essential that you follow the practical presentation attentively, replicate the analysis steps on the lab computer or your own laptop, and ask questions when necessary. You are encouraged to complete the exercises during and after practicals in groups. The practicals serve to start the exercises, and you are expected to complete them outside of the practical contact hours, alone or with others, and check your own learning progress. Do not rely on AI to solve the exercises as this will not support your learning.

8.3 Attendance and absences

You are expected to attend all lectures and practicals that are part of your programme. It is also expected that you arrive on time. Absence and late arrival are recorded on your university record. Inappropriate amounts of absence or late arrival at class, without extenuating circumstances, will be treated

seriously and may result in exclusion from the course. In addition, you should be aware that prospective employers almost always ask for information about attendance and punctuality, as well as matters such as your record on completing work to deadlines.

If you are unable to attend a session because of illness or other good reason you should notify the course lecturer/tutor and your Programme Administrator in advance (if possible). Absences of more than a few days should be backed up by medical or other evidence.

All absences will be reported to the relevant Tutor, who will then monitor your performance. A record of indifferent attendance will be held against you if your examination results are marginal; you should not expect to be shown sympathy by the Board of Examiners in such circumstances.

If you have missed a class, you should be sure to catch up on what you have missed by further independent reading of materials on the reading list and consulting any available lecture notes or slides if these are provided or asking other students whether they might allow you to consult theirs.

8.4 Drop-ins

Two drop-in sessions are scheduled in March. These sessions are led by the TA. Attendance is not mandatory. The drop-in sessions are meant as an opportunity to bring your questions and problems and have one-on-one or group interaction with the TA to get all your remaining questions answered and software problems solved.

8.5 Email and Canvas

Your commitment is also to check your university email and Canvas **at least every other day** – to make sure you are informed of any communications from tutors or administrative staff. These might concern, for example, important meetings with staff, changes of room; notification of course options, registration, or course-relevant information from your lecturer. Being unaware of arrangements because you have not checked your email or Canvas is not an acceptable excuse.

9 Assessment Criteria

Student's work will be assessed using the following criteria:

- **Structure and logic** of narration, coherence of the steps of analysis and interpretation.
- **Understanding** of key analytical notions, procedures, and measures.
- Quality of **justification of data presentation choices** with regard to important features of the data.
- Correctness and completeness of descriptive statistics calculation.
- Quality of **justification of model specification** with regard to key features of the data and taking into account previous analyses.
- Quality of **modelling**.
- Depth and correctness of **interpretation** given the specific research context.
- Comprehensiveness and formatting quality of **figures and tables**.
- Links to the **literature** on the research context (add to the mark but do not subtract from it).
- **Extra analytical/interpretation steps**, well-conducted and reasonable, – may add to the mark, but only in case mandatory requirements are met.
- Additional strengths/weaknesses not listed above.